

APPENDIX A CONVERGENCE ANALYSIS

In this section, we will demonstrate the convergence of iterative optimization by verifying that the optimization problem concerning $\mathbf{C}$ and $\mathbf{P}$ is convex.

Theorem 1. *Algorithm 1 provides the optimal solution with convexity.*

Proof. Assume we obtain $\mathbf{P}^t$ and $\mathbf{C}^t$ at the t -th iteration, and we set $\mathbf{C}$ as $\mathbf{C}^t$ to proceed to the next iteration of $\mathbf{P}$, which we formulate as $\mathbf{P}^{t+1}$. Let $\varphi(\mathbf{P})$ expressed as a standard quadratic form:

$$\begin{aligned}\varphi(\mathbf{P}) = & \frac{1}{2} \|\mathbf{H}\mathbf{S}^T \mathbf{P} - \mathbf{H}\mathbf{Y}\mathbf{C}\|_F^2 + \alpha \|\mathbf{P}\|_1 \\ & + \frac{1}{2} \lambda \text{tr}(\mathbf{S}^T \mathbf{P} \mathbf{L} \mathbf{P}^T \mathbf{S}),\end{aligned}\tag{A1}$$

where $\mathbf{L}$ is constructed from matrix $\mathbf{C}$ and $\mathbf{C}$ is a symmetric matrix with all values greater than zero. Thus, $\mathbf{L}$ is positive semi-definite and always satisfies $\mathbf{S}^T \mathbf{P} \mathbf{L} \mathbf{P}^T \mathbf{S} \geq 0$ for any nonzero $\mathbf{S}^T \mathbf{P}$. Therefore, we can confirm that $\text{tr}(\mathbf{S}^T \mathbf{P} \mathbf{L} \mathbf{P}^T \mathbf{S})$ is a convex function. It is commonly recognized that the standard ℓ_1 norm is convex, a property that is well known, and its convergence has been thoroughly investigated and proven in the literature. In addition, the remaining terms involving $\mathbf{P}$ exhibit convexity. Thus, the aggregation of three convex functions results in a convex optimization problem, which can be solved efficiently. Hence, we can obtain:

$$\varphi(\mathbf{P}^{t+1}) \leq \varphi(\mathbf{P}^t).\tag{A2}$$

Following the identical treatment, $\mathbf{P}$ is set to $\mathbf{P}^t$, and iterative solution for $\mathbf{C}^{t+1}$ is derived. Both Frobenius norms are convex, so we have:

$$\begin{aligned}& \frac{1}{2} \|\mathbf{H}\mathbf{S}^T \mathbf{P}^t - \mathbf{H}\mathbf{Y}\mathbf{C}^{t+1}\|_F^2 + \frac{1}{2} \gamma \|\mathbf{Y} - \mathbf{Y}\mathbf{C}^{t+1}\|_F^2 \\ & \leq \frac{1}{2} \|\mathbf{H}\mathbf{S}^T \mathbf{P}^t - \mathbf{H}\mathbf{Y}\mathbf{C}^t\|_F^2 + \frac{1}{2} \gamma \|\mathbf{Y} - \mathbf{Y}\mathbf{C}^t\|_F^2.\end{aligned}\tag{A3}$$

Combining Eqs. (A2) and (A3), the following inequality is obtained:

$$\begin{aligned}& \frac{1}{2} \|\mathbf{H}\mathbf{S}^T \mathbf{P}^{t+1} - \mathbf{H}\mathbf{Y}\mathbf{C}^{t+1}\|_F^2 + \frac{1}{2} \gamma \|\mathbf{Y} - \mathbf{Y}\mathbf{C}^{t+1}\|_F^2 \\ & \quad + \alpha \|\mathbf{P}^{t+1}\|_1 + \frac{1}{2} \lambda \text{tr}(\mathbf{S}^T \mathbf{P}^{t+1} \mathbf{L} (\mathbf{P}^{t+1})^T \mathbf{S}) \\ & \leq \frac{1}{2} \|\mathbf{H}\mathbf{S}^T \mathbf{P}^t - \mathbf{H}\mathbf{Y}\mathbf{C}^t\|_F^2 + \frac{1}{2} \gamma \|\mathbf{Y} - \mathbf{Y}\mathbf{C}^t\|_F^2 + \alpha \|\mathbf{P}^t\|_1 \\ & \quad + \frac{1}{2} \lambda \text{tr}(\mathbf{S}^T \mathbf{P}^t \mathbf{L} (\mathbf{P}^t)^T \mathbf{S}).\end{aligned}\tag{A4}$$

Eq. (A4) demonstrates the monotonic decrease of the objective function across iterations. We prove that Algorithm 1 yields a convex solution. Given that any local minimum in a convex problem is global, the proposed method achieves the global minimum as desired. $\square$

APPENDIX B COMPLEXITY AND RUNTIME ANALYSIS

The pseudo-code for the alternating optimization of M²FSAG is shown in Algorithm 1. In Algorithm 1, the maximum iteration times T is set to 100. The iteration stops when the change of the objective function between two consecutive iterations falls below 0.01 or when the maximum number of iterations is reached. The time complexity of M²FSAG in Algorithm 1 includes the following components: For Steps 1 and 2, the total computational complexity of the matrix transformation part is $O(ndK + nK)$, where n denotes the number of instances, d represents the number of features, K is the number of fuzzy rules. The time complexity of initializing matrix $\mathbf{C}$ in Step 3 is $O(l^2)$, where l represents the number of labels. For Step 4, the complexity of initializing matrices $\mathbf{P}_0$ and $\mathbf{P}_1$ is $O(K^2 nd^2 + K n l d)$. The complexity in step 6 is $O(l^2)$. The complexity of steps 7 and 9 is $O(1)$. The complexity in step 8 is $O(TKdn^2 + TK^2 d^2 n + TK^2 d^2 l + TK d n l + TK d l^2)$, where T denotes the number of iterations. Finally, the complexity of optimizing $\mathbf{P}$ is $O(ln^2 + n^2 l + lnKd + l^2 Kd)$, where the computational complexities for calculating $\mathbf{U}$ and $\mathbf{V}$ are $O(ln^2 + n^2 l)$ and $O(ln^2 + lnKd + l^2 Kd)$, respectively. Through the above discussion, we can discover that step 8 determines the overall time complexity. To simplify the results, let $a = \max(l, T, K)$ and $b = \max(n, Kd)$, where $a \leq b$ in general, the overall complexity of the entire method can be denoted as $O(ab^2(a + b))$.

Table A1 presents the average running times (in seconds) of ten algorithms on twelve datasets under five-fold cross-validation with a 30% missing rate. According to Table A1, we can obtain that the LSML and TSKFS achieve the shortest running times across most datasets, demonstrating high efficiency in multi-label learning tasks. In contrast, MLKNN and LFLI exhibit significantly longer computation times on large datasets, indicating limited scalability. On small datasets such as Image,

Algorithm 1 M²FSAG

Input: The feature matrix $\mathbf{X} \in \mathbb{R}^{d \times n}$ and label matrix $\mathbf{Y} \in \mathbb{R}^{l \times n}$; The number of fuzzy rules K ; The parameters α, γ , and λ ; The maximum iteration times T

Output: The consequent parameter matrix $\mathbf{P}$ of TSK fuzzy system;

- 1: Calculate membership using the Gaussian function according to Eq. (3);
 - 2: Convert the input matrix $\mathbf{X}$ into fuzzy matrix $\mathbf{S}$;
 - 3: Initialize the adaptive graph matrix $\mathbf{C}$ randomly;
 - 4: $\omega_0, \omega_1 \leftarrow 1, \mathbf{P}_0, \mathbf{P}_1 \leftarrow (\mathbf{S}\mathbf{S}^T + \lambda_1 \mathbf{I})^{-1} \mathbf{S}\mathbf{Y}^T$, and $t \leftarrow 1$;
 - 5: **while** not converged **do**
 - 6: Update the graph Laplacian matrix $\mathbf{L}$ by $\mathbf{C}$;
 - 7: $\mathbf{P}^{(t)} \leftarrow \mathbf{P}_t + ((\omega_{t-1} - 1)/\omega_t)(\mathbf{P}_t - \mathbf{P}_{t-1})$;
 - 8: $\mathbf{Z}^{(t)} \leftarrow \mathbf{P}^{(t)} - \frac{1}{L_f} f'(\mathbf{P}^{(t)})$;
 - 9: Update $\mathbf{P}_{t+1}$ based on the expression in Eq. (26);
 - 10: Compute $\mathbf{C}$ according to Eq. (32);
 - 11: $\omega_{t+1} \leftarrow (1 + \sqrt{4\omega_t^2 + 1})/2$;
 - 12: $t \leftarrow t + 1$;
 - 13: **end while**
-

LSML and TSKFS complete within 0.27 and 0.47 seconds respectively, while M²FSAG requires 4.35 seconds. Nevertheless, M²FSAG is still faster than SMDKM and TLNM on large datasets. This efficiency is primarily due to its iterative mechanism combining nonlinear modeling with adaptive graph learning. Although multiple iterations are involved, M²FSAG maintains moderate running times on small and medium datasets and exhibits clear advantages on large-scale datasets. For large datasets such as Bibtex and TMC2007-500, M²FSAG substantially reduces computation time while maintaining robust performance, demonstrating excellent scalability. Compared to TLNM, which also employs nonlinear modeling, M²FSAG achieves superior results across most datasets, further validating the effectiveness of using TSK fuzzy systems to capture nonlinear relationships between features and labels. Overall, LSML and TSKFS are the fastest algorithms and suitable for applications prioritizing efficiency. M²FSAG achieves a good balance of efficiency and performance on medium and large datasets, significantly outperforming MSWL and TLNM, highlighting the advantage of combining nonlinear modeling with adaptive graph learning.

TABLE A1
THE AVERAGE RUNNING TIMES OF THE TEN ALGORITHMS ON EACH DATASET (IN SECONDS).

Dataset	MLKNN	NFDML	LSML	MSWL	SMDKM	LFLI	TLNM	TSKFS	HDTSK	M ² FSAG
Bibtex	1894.66	1116.51	7.86	1307.29	157.96	238.91	520.70	15.26	779.21	153.07
Business	56.99	86.86	0.96	156.93	49.50	59.82	97.53	2.08	51.85	73.20
CAL500	1.72	3.92	0.34	178.92	11.02	0.68	42.58	0.33	21.10	49.74
Computer	95.16	95.82	1.21	467.93	49.80	56.40	85.88	4.92	92.27	35.77
Core15k	80.00	228.56	5.85	1086.60	138.08	78.30	570.03	13.43	181.92	108.80
Education	74.41	115.86	1.13	146.24	60.56	50.53	109.17	3.28	80.46	147.34
Image	6.22	16.79	0.27	14.39	17.93	3.50	27.28	0.47	9.92	4.35
Recreation	80.17	63.17	1.00	551.76	56.17	44.52	111.34	3.52	77.71	28.44
Reference	124.16	182.68	1.31	78.94	53.84	56.13	123.15	6.67	111.75	49.71
Science	104.27	159.97	1.38	320.00	53.70	53.87	133.43	6.05	107.33	44.58
Society	87.23	87.27	1.00	54.93	49.09	55.52	78.09	3.99	76.36	16.51
TMC2007-500	2333.80	697.20	4.53	592.36	372.28	1821.05	467.29	6.51	230.54	103.25

APPENDIX C
EXPERIMENTAL RESULTS

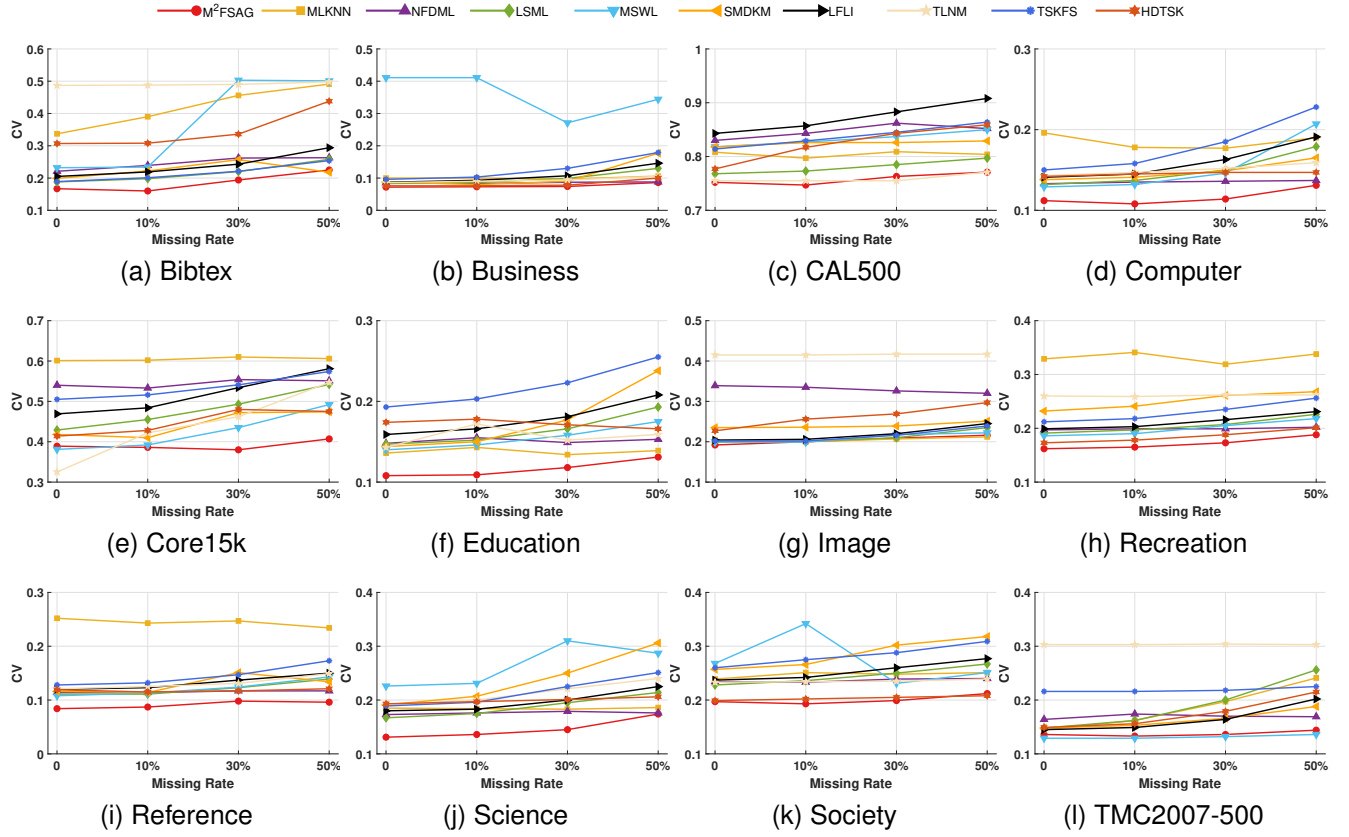


Fig. A1. CV performance comparison of all algorithms under different missing rates (0%, 10%, 30%, and 50%) on 12 datasets. Lower CV indicates better performance. M²FSAG achieves comparable or superior results on most datasets, especially when the missing rate is high.

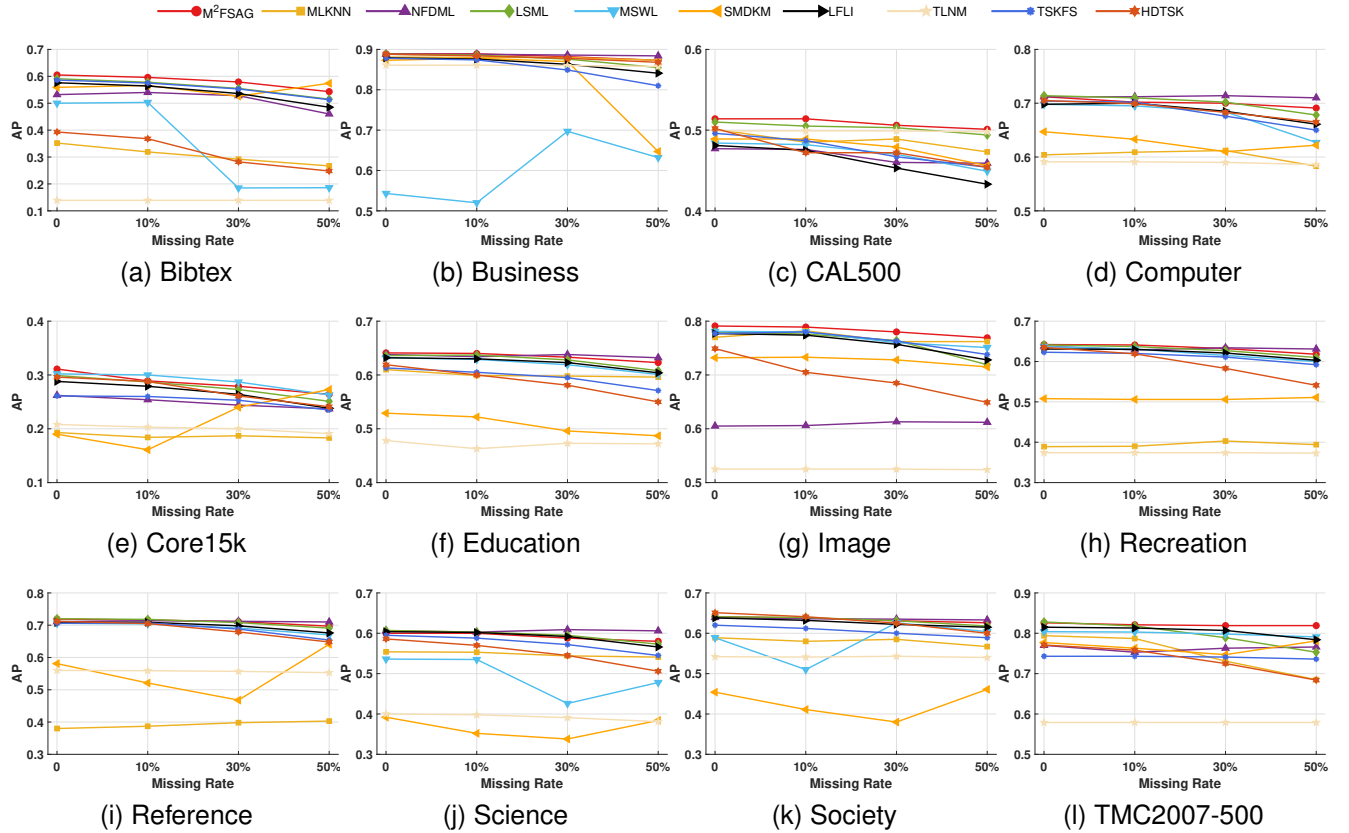


Fig. A2. AP performance comparison of all algorithms under different missing rates (0%, 10%, 30%, and 50%) on 12 datasets. Higher AP indicates better performance. M²FSAG achieves comparable or superior results on most datasets, especially when the missing rate is high.